

F

FLUTTER COURSE

Session 1 Assignment

Flutter Basics & Dart Fundamentals

Instructor

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Total Marks

100 (+5 Bonus)

Instructions

- Write all answers yourself.
- Do not copy from classmates.
- Use Dart language.
- Write clean and readable code.
- Try every question before asking for help.
- Submit both your Dart code and screenshots of the output.
- Total Marks: 100 (+5 Bonus)

Complete every section in order. Sections progress from easy to more challenging. Blank spaces are provided for handwritten answers and code — use extra paper if needed.

1. What is Mobile Development?

Creating software applications to run on mobiles

2. What is Native Development?

Building or developing a mobile application that works only on one specific platform

3. What is Cross Platform?

Building or developing a mobile application that works on different platforms, adapting to its style and users

4. Mention three Cross Platform frameworks.

1. Flutter 2. React Native 3. Kotlin

5. What are the native languages for Android and iOS?

Android uses Kotlin, iOS uses Swift

6. Explain one advantage of Flutter.

Flutter uses widgets; it draws every pixel on the screen

7. Explain one difference between Flutter and React Native.

React Native adapts its style to the platform it's on, so the UI changes from platform to another.

Flutter, however, draws its own style so the UI is unified on all platforms

Write the correct Dart data type for each value.

1.

5

Answer: int

2.

3.14

Answer: double

3.

"Kareem"

Answer: String

4.

true

Answer: boolean

5.

["Ahmed", "Ali"]

Answer: list

6.

```
{  
  "name": "Ahmed"  
}
```

Answer: map

Write Dart code for each of the following.

1. Create an integer variable called age.

Write your Dart code here

```
int age= 21;
```

2. Create a String variable called studentName.

Write your Dart code here

```
String studentName= "Malak";
```

3. Create a bool variable called isStudent.

Write your Dart code here

```
bool isStudent= true;
```

4. Create a double variable called salary.

Write your Dart code here

```
double salary= 10000.00;
```

5. Create a List of three names.

Write your Dart code here

```
List names= ["Malak", "Waad", "Hanadi"];
```

6. Create a Map containing: *name*, *age*, *city*

Write your Dart code here

```
Map dict= {  
  "name" : "Malak",  
  "age" : 21,  
  "city" : "Cairo"  
};
```

1. What is the difference between var and dynamic?

var allows you to input any value of any data type, but after one data type is used, you can't change it to another datatype later on. dynamic on the other hand, allows the change of its data type whenever needed during compile time

2. When should you use dynamic?

when you don't know which data type to use or if you plan on changing the input value later on with another one of different data type

3. Write one example using var.

Write your Dart code here

```
var value1 = 72;  
//value1= "duck"; //(gives error message)
```

4. Write one example using dynamic.

Write your Dart code here

```
dynamic value2 = 48;  
//value2= true; //(works)
```

1. What is the difference between final and const?

final is added before any input to prevent any changes to it during compile time, but

changes can be added during runtime. const doesn't allow any changes to its values,

neither in compile time nor in runtime

2. When is final evaluated?

During runtime only

3. When is const evaluated?

It's written and fixed during compile time and never changes

4. Write one example using final.

Write your Dart code here

```
final List l1 = [3, 4, 5];
```

5. Write one example using const.

Write your Dart code here

```
const List l2 = [5, 6, 7];
```

Write Dart code only for each instruction below.

1. Create a List of five cities.

Write your Dart code here

```
final List<String> cities= ["Cairo", "Alexandria", "Aswan", "Luxor", "Hurghada"];
```

2. Add one city.

Write your Dart code here

```
cities.add("Sinai");
```

3. Remove the third city.

Write your Dart code here

```
cities.removeAt(3);
```

4. Insert a city at index 2.

Write your Dart code here

```
cities.insert(2, "Asiut");
```

5. Print the first item.

Write your Dart code here

```
print(cities.first);
```

6. Print the last item.

Write your Dart code here

```
print(cities.last);
```

7. Print the length of the List.

Write your Dart code here

```
print(cities.length);
```


8. Check if the List contains "Cairo".

Write your Dart code here

```
print(cities.contains("Cairo"));
```

9. Sort the List.

Write your Dart code here

```
cities.sort();
```

10. Clear the List.

Write your Dart code here

```
cities.clear();
```

Write Dart code using each of the following functions.

1. Use map()

Write your Dart code here

```
L1.map((n) => n * 2);
```

2. Use where()

Write your Dart code here

```
L1.where((n) => n % 2 == 0);
```

3. Use any()

Write your Dart code here

```
L1.any((n) => n >= 5);
```

4. Use every()

Write your Dart code here

```
L1.every((n) => n >= 4);
```

5. Use forEach()

Write your Dart code here

```
L1.forEach(print);
```

1. Create a Student Map containing: *name*, *age*, *grade*

Write your Dart code here

```
Map<String, dynamic> student= {  
  "name" : "Malak",  
  "age" : 21,  
  "grade" : "97"  
};
```

2. Print the student's name.

Write your Dart code here

```
print(student["name"]);
```

3. Update the age.

Write your Dart code here

```
student["age"] = 22;
```

4. Add a new key called address.

Write your Dart code here

```
student["address"] = "new nozha";
```

5. Remove grade.

Write your Dart code here

```
student.remove("grade");
```

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Runtime vs Compile Time

5 Marks

1. What is Compile Time?

it is the code when it is compiled, before the app runs

2. What is Runtime?

it is when the app runs

B

Coding Challenge (Bonus)

5 Marks

Without looking at your notes, write a complete Dart program that demonstrates ALL of the following:

- | | | |
|-------------------|----------|-------------------|
| ✓ Variables | ✓ int | ✓ double |
| ✓ bool | ✓ String | ✓ List |
| ✓ Map | ✓ var | ✓ dynamic |
| ✓ final | ✓ const | ✓ List operations |
| ✓ Printing output | | |

Write your code below.

Write your complete Dart program here

Challenge Question (Bonus)

Bonus Marks

Create a Student Management program that:

- Stores three students inside a List.
- Each student is represented as a Map.
- Print every student's name.
- Add a new student.
- Remove one student.
- Print the final List.

(Write the complete code.)

Write your complete Dart program here